

Qn. Define Layers of a Deep Neural Network, Back propagation & Forward propagation?

Artificial Neural Networks (ANNs) are inspired by the structure and functioning of the human brain. When these networks contain multiple hidden layers, they are referred to as **Deep Neural Networks (DNNs)**. DNNs are the backbone of modern artificial intelligence applications such as image recognition, natural language processing, and autonomous systems. To understand how they work, it is essential to study their **layers**, the process of **forward propagation**, and the learning mechanism called **backpropagation**.

Layers of a Deep Neural Network

A DNN is organized into distinct layers, each serving a specific role in transforming input data into meaningful output.

- **Input Layer**
 - The entry point of raw data into the network.
 - Each neuron corresponds to one feature of the dataset. For example, in image classification, each pixel value is an input neuron.
 - No computation occurs here beyond passing the data forward.
- **Hidden Layers**
 - These are the intermediate layers between input and output.
 - Each hidden layer applies a linear transformation followed by a non-linear activation function.
 - The linear transformation is expressed as:

$$z = W \cdot x + b$$

where W are weights, x is the input vector, and b is the bias.

- The activation function (e.g., ReLU, Sigmoid, Tanh) introduces non-linearity, enabling the network to learn complex patterns.
- With multiple hidden layers, the network can progressively extract higher-level features. For instance, in image recognition:
 - Early layers detect edges.
 - Middle layers detect shapes or textures.
 - Deeper layers detect entire objects.
- **Output Layer**
 - Produces the final prediction.
 - The choice of activation depends on the task:
 - **Softmax** for multi-class classification.
 - **Sigmoid** for binary classification.

- **Linear activation** for regression tasks.

- The output is compared with the true target to compute the error or loss.

Forward Propagation

Forward propagation is the **process of passing data through the network from input to output.**

- **Step-by-Step Process**

1. Input data enters the input layer.
2. Each hidden layer computes weighted sums and applies activation functions.
3. The transformed data flows sequentially through all hidden layers.
4. The output layer produces predictions.

- **Purpose** Forward propagation computes the **predicted output** given the current parameters. It is essentially the “guess” of the network before learning adjustments are made.

Backpropagation

Backpropagation is the **learning algorithm** that allows the network to improve its predictions by updating weights and biases.

- **Step-by-Step Process**

1. **Loss Calculation**

- The difference between predicted output and actual target is measured using a loss function (e.g., Mean Squared Error, Cross-Entropy).

2. **Gradient Computation**

- Using the chain rule of calculus, the gradient of the loss with respect to each weight is calculated.
- This tells us how much each parameter contributed to the error.

3. **Backward Flow**

- Gradients are propagated backward from the output layer to the input layer.
- Each layer receives error information and computes its own gradient.

4. **Weight Update**

- Parameters are updated using optimization algorithms such as Gradient Descent:

- **Purpose** Backpropagation ensures that weights are adjusted in the direction that reduces error, allowing the network to learn patterns more accurately over multiple training iterations (epochs).

Relationship Between Forward and Backpropagation

- **Forward Propagation:** Data flows forward, predictions are made.

- **Backpropagation:** Errors flow backward, parameters are updated.
- Together, they form the **training cycle** of a neural network. Each iteration improves the model's ability to generalize and make accurate predictions.